

SupplyGuard AI

Supply Chain Analytics & Risk Intelligence Dashboard

IBM SkillsBuild — Data Analytics with AI Internship

Project Report

1. Executive Summary

SupplyGuard AI is an interactive, end-to-end supply chain analytics and risk intelligence dashboard developed using Python and Streamlit. The project was built as part of the IBM SkillsBuild Data Analytics with AI Internship programme, with the goal of transforming 180,519 raw supply chain order records into clear, actionable business intelligence.

The dashboard covers the complete data analytics pipeline — data overview, cleaning validation, statistical exploration, delivery and shipping analysis, sales and profit analysis, product performance analysis, risk and anomaly detection, and AI-driven narrative insights. All analysis is presented in a clean white-and-blue web interface with colour-coded KPI cards, multi-colour charts, and natural-language insight summaries.

Key headline findings from the analysis include: total revenue of \$36.78M over four years, a critically high late delivery rate of 57.28%, 33,784 loss-making orders representing 18.71% of all transactions, 4,062 fraud-flagged orders, and \$3.88M in revenue currently at risk. These findings point to clear, prioritised areas for operational improvement.

2. Project Overview

Project Name: SupplyGuard AI

Full Title: Supply Chain Analytics & Risk Intelligence Dashboard

Programme: IBM SkillsBuild — Data Analytics with AI Internship

Domain: Supply Chain Management / Business Analytics

Focus Areas: Orders, Sales, Profit, Late Deliveries, Fraud Detection, Risk Intelligence

The primary objective of SupplyGuard AI is to provide supply chain managers, business analysts, and operations teams with a self-service analytics tool that surfaces risks, identifies underperforming areas, and delivers clear recommendations — all without requiring SQL or programming knowledge. The entire pipeline from data to insight is contained in a single Streamlit application file (app.py).

3. Dataset Description

Source Dataset: DataCo Smart Supply Chain for Big Data Analysis (DataCoSupplyChainDataset.csv). The dataset is a publicly available, real-world supply chain transaction dataset widely used for academic and analytical research.

The cleaned and feature-engineered version (supply_chain_cleaned.csv) is used by the application. It contains 180,519 rows and 62 columns, including 22 engineered features.

Attribute	Detail
Total Records	180,519 rows
Total Features	62 columns
Unique Orders	65,752
Unique Products	118
Product Categories	50
Countries Covered	164
Departments	11 (Fitness, Apparel, Golf, Footwear, Outdoors, Fan Shop, Technology, Book Shop, Discs Shop, Pet Shop, Health and Beauty)
Markets	5 (Europe, USCA, LATAM, Pacific Asia, Africa)
Customer Segments	Consumer, Corporate, Home Office
Shipping Modes	Standard Class, First Class, Second Class, Same Day
Years Covered	2015, 2016, 2017, 2018
Payment Types	DEBIT, TRANSFER, CASH, PAYMENT

3.1 Engineered Features

The following features were engineered from the raw data during the cleaning phase:

- `is_delayed` — Binary flag (1/0) indicating whether the actual shipping days exceeded the scheduled days.
- `shipping_delay_days` — Numeric difference between actual and scheduled shipping days.
- `fulfillment_days` — Total calendar days from order placement to shipment.
- `is_loss_order` — Binary flag (1/0) identifying orders where `order_profit` is negative.
- `is_fraud_flagged` — Binary flag combining multiple fraud signals into a single alert indicator.
- `fraud_risk_score` — Composite score from 0 to 4 based on fraud signal combinations.
- `revenue_at_risk` — Revenue value exposed to loss from late-delivered or cancelled orders.
- `profit_margin_pct` — Calculated as $(\text{order_profit} / \text{sales}) \times 100$.
- `discount_impact` — Product of discount rate and sales value, measuring absolute discount cost.
- `is_transfer_payment` — Binary flag isolating orders paid via TRANSFER method for fraud correlation.
- `order_year`, `order_month`, `order_quarter` — Temporal decomposition of the order date for trend analysis.
- `year_month` — YYYY-MM string label for monthly trend charts and time-series grouping.

4. Technology Stack

SupplyGuard AI is built entirely in Python, using a minimal and reproducible dependency set. The full stack is listed below:

Library	Version	Purpose
streamlit	≥ 1.32.0	Web application framework — UI, sidebar, tabs, widgets, layout
pandas	≥ 2.0.0	Data loading, cleaning, groupby aggregation, pivoting, filtering
numpy	≥ 1.26.0	Numerical computations, array operations, statistical helpers
matplotlib	≥ 3.8.0	All charts — bar, horizontal bar, line with fill, pie, scatter, histogram
warnings (stdlib)	built-in	Suppresses non-critical runtime warnings for clean output

The application runs as a single-file Streamlit app (app.py) with the cleaned dataset (supply_chain_cleaned.csv) in the same directory. No database, API keys, or external services are required.

5. Analytics Pipeline

The project follows a structured, eight-stage data analytics pipeline. Each stage maps to a specific section of the dashboard:

- Stage 1 — Overview KPI Dashboard: Presents 8 headline KPIs giving a full business health snapshot. Metrics include total orders, revenue, profit, average margin, late delivery rate, fraud-flagged count, loss order count, and total revenue at risk. Supporting charts show annual revenue, delivery status split, market order volume, and annual profit.
- Stage 2 — Cleaned Dataset Validation: Displays row/column counts, missing value totals, duplicate detection, per-column data types, null counts, and unique value counts. A 10-row sample record preview is also shown.
- Stage 3 — Data Exploration: Provides descriptive statistics for 8 key numeric columns, interactive distribution histograms with gradient colouring, and pattern analysis by customer segment, market, shipping mode, order status, and payment type.
- Stage 4 — Delivery and Shipping Analysis: Analyses on-time vs. late delivery rates across segments, markets, shipping modes, and regions. Includes monthly trend charts and yearly performance breakdown tables.
- Stage 5 — Sales and Profit Analysis: Covers annual and monthly revenue and profit trends, segment-level profitability, market and regional revenue comparisons, and the impact of discount rates on profit margin using a scatter plot and distribution analysis.
- Stage 6 — Product Analysis: Identifies the top 10 products by revenue and profit, analyses performance by department and category, and surfaces loss leader products and those with the highest loss-order rates.

- Stage 7 — Risk and Anomaly Analysis: Quantifies fraud exposure by payment type and market, analyses the fraud risk score distribution, tracks loss order rates by department and shipping mode over time, and maps revenue at risk by market and shipping mode.
- Stage 8 — AI Insights: Converts all quantitative analysis into eight natural-language insight cards with colour-coded borders, a summary scorecard table, and a business health metrics bar chart comparing key operational rates.

6. Dashboard Sections

The application is organised into six sidebar sections, accessible via a radio navigation menu:

Section	Icon	Content Summary
Overview & Data		8 KPI cards, 4 overview charts, cleaned dataset info, data exploration tabs
Delivery & Shipping		4 delivery KPIs, delay overview, shipping mode analysis, market/region, monthly trends
Sales & Profit		4 financial KPIs, annual trends, segment profitability, market revenue, discount analysis
Product Analysis		Top 10 by revenue/profit, department/category breakdown, loss leaders
Risk & Anomaly		Fraud rate by payment/market, fraud score distribution, loss orders, revenue at risk
AI Insights		8 insight cards, summary scorecard table, business health metrics chart

7. Key Findings & Insights

The following findings are drawn directly from the data analysis performed in the dashboard. All numbers are computed from the supply_chain_cleaned.csv dataset.

7.1 Financial Performance

Metric	Value
Total Revenue (2015–2018)	\$36,784,735
Total Net Profit	\$3,966,903
Average Profit Margin	10.83% per order item
Loss-Making Orders	33,784 orders (18.71% of total)
Total Revenue at Risk	\$3,883,547
Fraud-Flagged Orders	4,062 orders (2.25% of total)
Top Revenue Market	Europe
Best Dept by Profit	Fan Shop

7.2 Delivery Performance

The most critical operational issue revealed by the data is the delivery performance. With a late delivery rate of 57.28%, more than half of all shipments in the dataset

were delivered after the scheduled date. This is the highest-priority risk for the business:

- On-time delivery rate: only 17.84% — fewer than 1 in 5 orders arrive on schedule.
- Late delivery rate: 57.28% across all markets.
- Advance shipping (early): 20.58% of orders — showing the shipping schedule has room.
- Cancelled shipments: 4.30% of orders cancelled before delivery.
- Average delay when late: 1.62 days per delayed shipment.
- Worst-performing market by late delivery rate: Europe.
- Most delay-prone shipping mode: First Class (highest late delivery rate despite premium positioning).

7.3 Sales and Segment Insights

The Consumer segment is the largest by order volume and revenue, followed by Corporate and Home Office. Europe is the top market by total revenue, though it also holds the highest late delivery rate — indicating high-volume/high-risk exposure. The Fan Shop department outperforms all others in net profit, while the Book Shop consistently produces losses.

- Top revenue product: Field & Stream Sportsman 16 Gun Fire Safe.
- Best-performing department: Fan Shop (highest net profit).
- Worst-performing department: Book Shop (net loss across the period).
- Discount-margin correlation: -0.023 (negative). Higher discounts correlate with lower margins.
- Orders with discount rates exceeding 10% are disproportionately represented in loss-making transactions.

8. Risk & Fraud Analysis

Risk analysis is one of the core differentiators of SupplyGuard AI. The dashboard identifies and quantifies three distinct categories of business risk:

8.1 Fraud Risk

4,062 orders (2.25% of all orders) are flagged as potentially fraudulent by the `is_fraud_flagged` indicator. The `fraud_risk_score` (0–4) provides a tiered severity view. Key findings:

- TRANSFER payment method has the highest fraud association rate of any payment type.
- Orders with SUSPECTED_FRAUD status in `order_status` confirm the flag accuracy.
- The dashboard surfaces all orders with `fraud_risk_score` ≥ 2 for direct operational review.
- Fraud exposure by market is distributed across all 5 global markets, with no single market dominating — indicating a systemic risk rather than a geographic one.

8.2 Loss Order Risk

33,784 orders (18.71%) are loss-making — meaning the business sold below cost or at a profit margin below zero. This is an exceptionally high loss rate for a commercial operation. Contributing factors identified:

- Aggressive discounting: The Book Shop and Discs Shop departments show the highest loss-order rates, both heavily discount-driven.
- Shipping cost absorption: Standard Class shipping, despite being the lowest tier, shows a notable loss rate — suggesting shipping cost overruns on small-value orders.
- Monthly trend: Loss order rate shows no improving trend over the 2015–2018 period, indicating the issue is structural rather than temporary.

8.3 Revenue at Risk

\$3.88M in total revenue is classified as at-risk. This figure represents the combined revenue from late-delivered and loss-making orders where the business may not recover full value. Europe and LATAM are the markets with the highest at-risk revenue. Standard Class and First Class shipping modes carry the greatest portion of at-risk revenue by volume.

9. Recommendations

Based on the full analysis performed across all eight dashboard sections, the following six recommendations are prioritised by impact:

- Recommendation 1 — Discount Policy Reform: Implement a maximum discount cap of 10% across all segments. Orders exceeding this threshold are significantly more likely to generate losses. A controlled discount authorisation workflow should be introduced for any exception above the cap.
- Recommendation 2 — Delivery Performance Overhaul: The 57.28% late delivery rate is the single most damaging operational metric. A dedicated task force should audit the end-to-end shipping workflow for Europe and First Class shipments specifically, as these show the worst performance despite premium expectations.
- Recommendation 3 — Fraud Prevention System: Implement automated real-time screening for TRANSFER payment orders, particularly those with `fraud_risk_score` ≥ 2 . A manual review queue should be created for score-3 and score-4 orders before fulfilment is approved.
- Recommendation 4 — Product Portfolio Review: Conduct a full profitability review of the Book Shop and Discs Shop departments. Products with a loss-order rate above 40% should be either repriced, discontinued, or subject to a minimum-margin rule before being offered with discounts.
- Recommendation 5 — Predictive Delivery Analytics: Build a machine learning classifier using the existing engineered features (`late_delivery_risk`, `shipping_mode_enc`, `market_enc`, `fulfillment_days`) to predict late deliveries before dispatch. This would enable proactive customer communication and rerouting decisions.

- Recommendation 6 — Monthly Management Reporting: SupplyGuard AI should be adopted as the standard monthly reporting tool for regional supply chain managers. Segment-level KPI dashboards should be reviewed at the start of each month, with specific attention to the Revenue at Risk metric as a leading indicator.

10. Conclusion

SupplyGuard AI successfully demonstrates the end-to-end application of the data analytics pipeline to a real-world supply chain dataset. Starting from 180,519 raw transaction records, the project delivers a complete, self-contained analytics platform that is accessible without programming expertise, readable on any device, and immediately actionable by supply chain and finance teams.

The dashboard surfaces four critical business problems that may not have been visible through standard reporting: a majority late-delivery rate masking systemic logistics failure, a structurally high loss-order rate driven by uncapped discounting, quantifiable fraud exposure concentrated in transfer payments, and significant revenue at risk that grows month-over-month without intervention.

The project was built in adherence to the IBM SkillsBuild Data Analytics with AI Internship programme objectives, demonstrating proficiency in data cleaning, exploratory data analysis, domain-specific analytics, risk detection, and the communication of findings through an interactive visual application.

All source code, the cleaned dataset, this project report, the requirements file, and the README documentation are included in the project submission package.

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